Creating and Deploying Production-Ready Agentic Systems

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Part I

Autonomous LLM Agents

What we will be talking about

1. The LLM as a System Component
   - Inherent LLM limitations
   - When to use LLMs?
2. Tool-Based Reasoning using ReAct
   - Thought, Action, Observation
   - Standardization
3. Agent-to-Agent Interaction
   - Single-Agent vs. Multi-Agent
   - Orchestrating Interaction
   - What do we actually need?

Outline

Section 1

The LLM as a System Component

Subsection 1

Inherent LLM limitations

Mathematics

LLMs are really bad at math

- Predict the most likely sequence of tokens, not the logically necessary derivation.
- Probabilistic nature → prone to hallucinations.
- Tokenization absolutely destroys numbers.

While OK for small numbers (they can memorize some patterns), they generally fail at larger ones.

Word counting

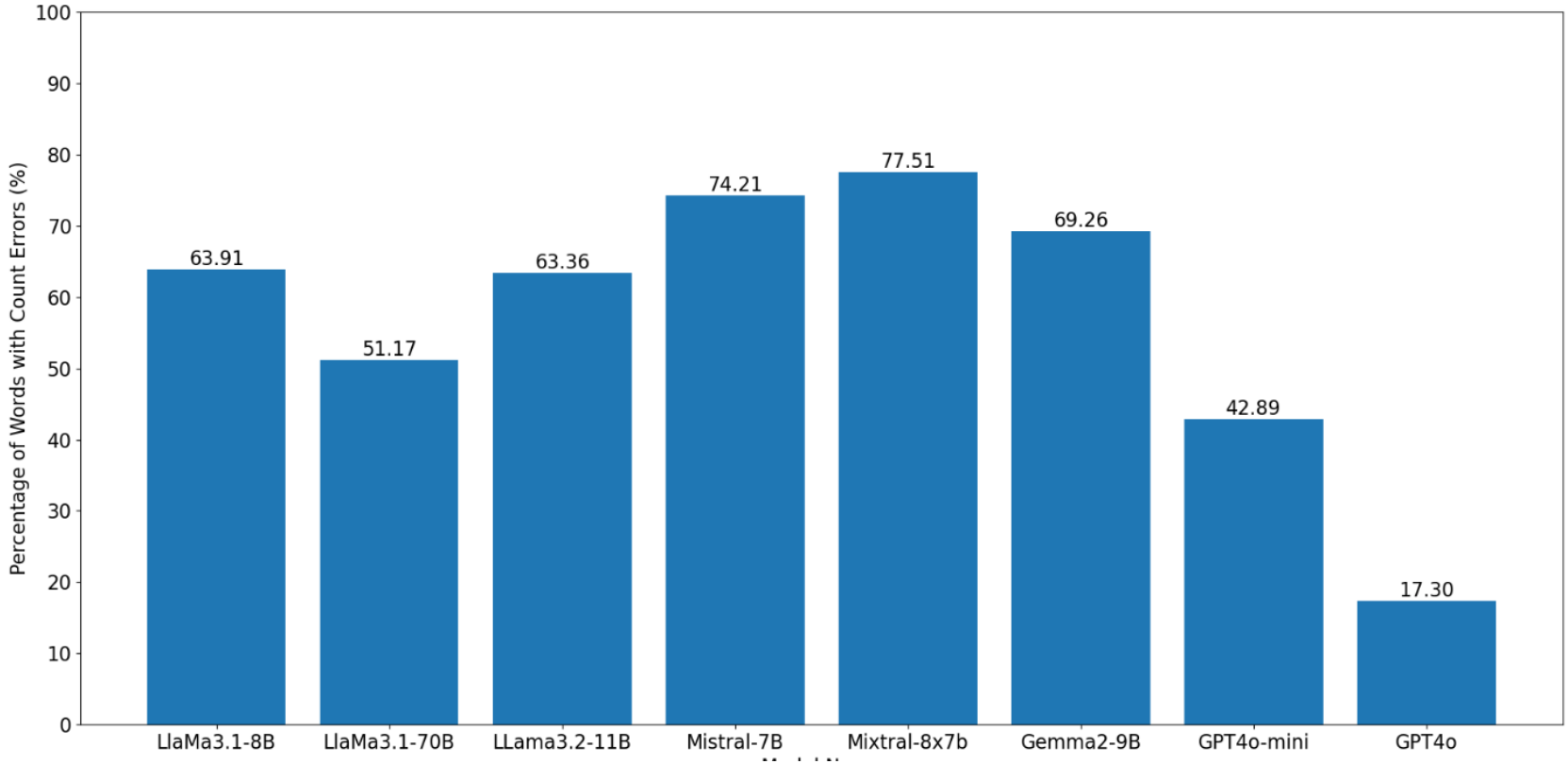


Figure: Percentage of words with errors on when counting letters for the different models https://arxiv.org/html/2412.18626v1.

Character counting

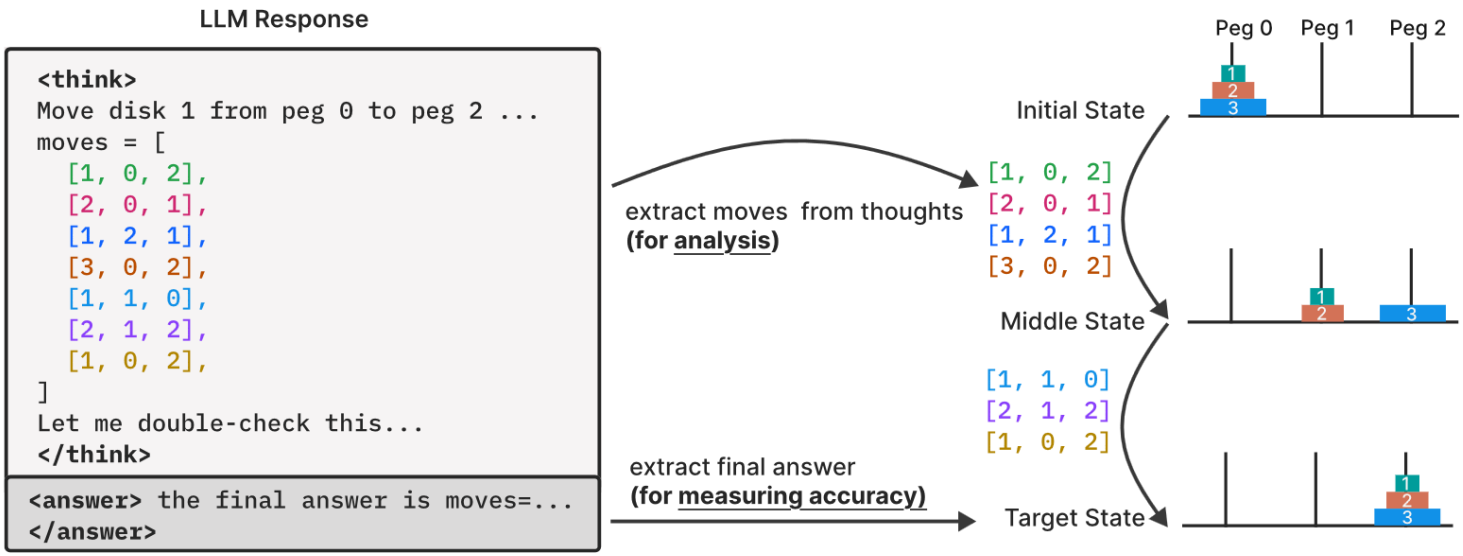
Example

model.prompt("How many 'r's are in 'carr'?")
>>> "There is one 'r' in 'carr'".

Indicative explanation

- The model can not find the word "carr" in its knowledge base.
- But it knows the word "car" and the fact that "carr" and "car" are highly correlated.
- "car" has one "r", therefore this is the answer.

Deductive reasoning



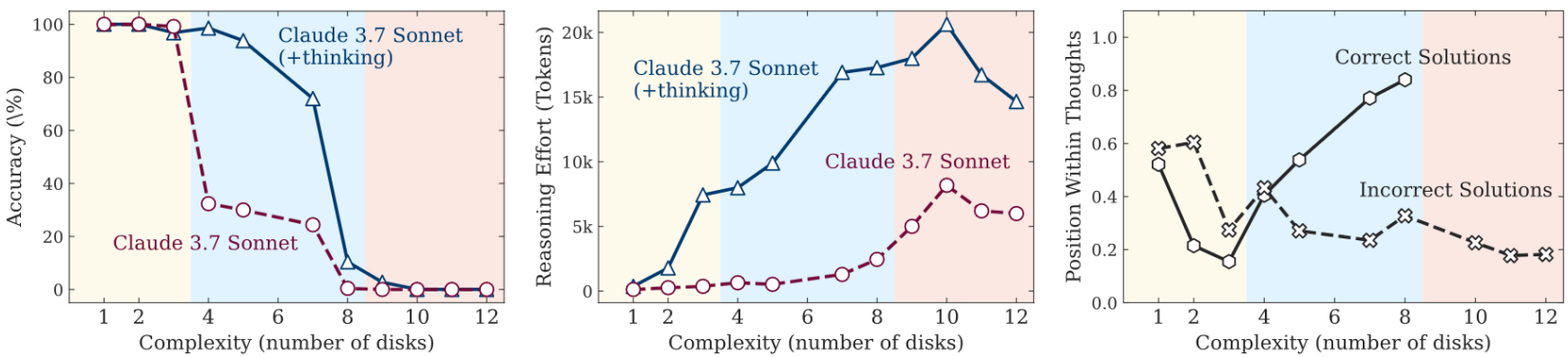


Figure: Percentage of words with errors on when counting letters for the different models https://arxiv.org/html/2412.18626v1.

General Limitations

LLMs cannot:

- Think.
- Behave.
- Deduct.

Why?

- LLMs are generally excellent stochastic parrots. Their next-word-prediction nature prevents them from understanding the world and its mechanisms.
- Finetuning, reasoning and in-context learning alleviate, but do not bypass this restriction.

A Note on Benchmarks

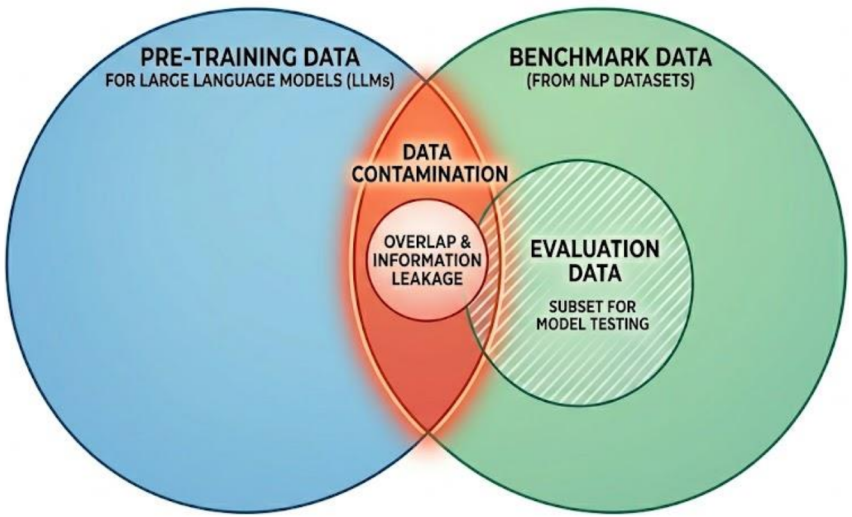
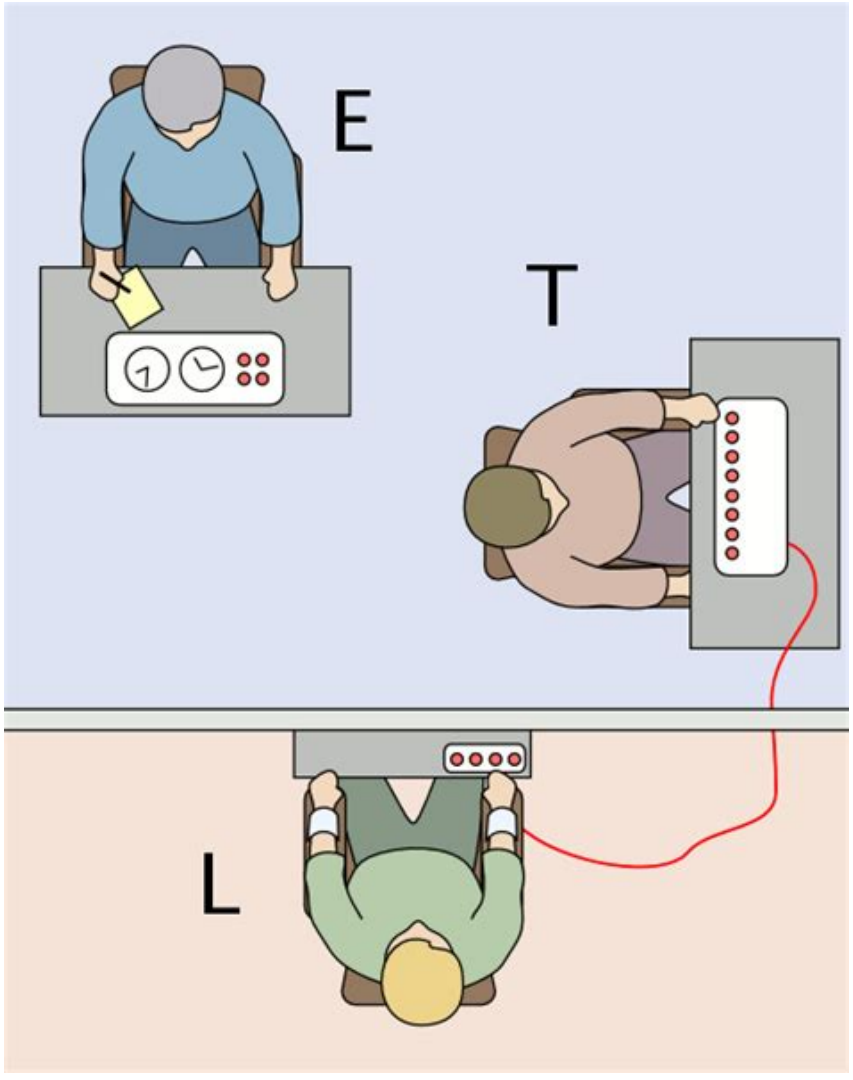


Figure: Credit to Electra Maria Papadopoulou—generated with Gemini.

Data Contamination

- Evaluation data appears in training.
- Web-scraping may include datasets used as training, validation, or benchmarks, unknowingly inflating reported performance.
- Sometimes on purpose ("benchmark-maxxing").

A Note on Benchmarks



Behavior replication

- LLMs replicate the Milgram study.
- Emergent social property!
- ...or is it that the LLMs already know the results of the study?
- Perhaps, they implicitly follow what they know humans do.

A Note on Benchmarks

Benchmaxxing

Optimizing a LLM's performance primarily for high scores on public leaderboards and benchmarks, rather than practical, and reliable real-world capability.

How?

- Training Data Contamination.
- Format Overfitting.
- Checkpoint Cherry-Picking.
- Prompt Engineering.

Section 1

The LLM as a System Component

Subsection 2

When to use LLMs?

LLMs are not just task-specific tools

- So far, we have used LLMs as specialized tools for specific tasks.
- While useful, these tasks can generally be accomplished by simpler and more scalable models (with some performance loss).
- Where LLMs differ, is that they are autonomous, which allows for previously impossible use-cases.

When you are holding a hammer...

- Suppose you are given the following task, with appropriate resources from management.

Use case

Find all reviews that concern competitors of our product, and tell us what percentage of them mention it.

When you are holding a hammer...

Our problem could be decomposed in three parts

1. Discovery: Find pages containing reviews.
2. Filtering: Find the HTML sections containing the reviews.
3. Analysis: Find which reviews reference the product.

Which of these must necessarily be solved with LLMs? Which can be solved with traditional ways? What trade-offs are we making in the latter case?

... you treat everything as if it were a nail

The easy solution

We instruct an LLM to retrieve reviews via RAG. We isolate the reviews using prompting, then instruct the LLM to find whether our product is mentioned.

Result

Code is deployed within the day. Management is billed 50,000€ in a week.

But maybe we can do better

The system solution

We use a traditional scrapper to find relevant reviews. For each page, we instruct the LLM to extract the reviews. We then run reviews.str.contains(product_name).sum().

Result

Code is deployed within two days. Management is billed 20€ in a week.

Outline

Section 2

Tool-Based Reasoning using ReAct

Subsection 1

Thought, Action, Observation

Tool-Calling

Tool example

```
from smolagents import tool

@tool
def calculator(expr: str) -> float:
    """
    Evaluate a basic arithmetic expression.
    Args:
    expr: The math expression to evaluate, such as "2 + 3 * (4 / 2)".
    """
    return eval(expr, {"__builtins__": None}, {})
```

What is ReAct?

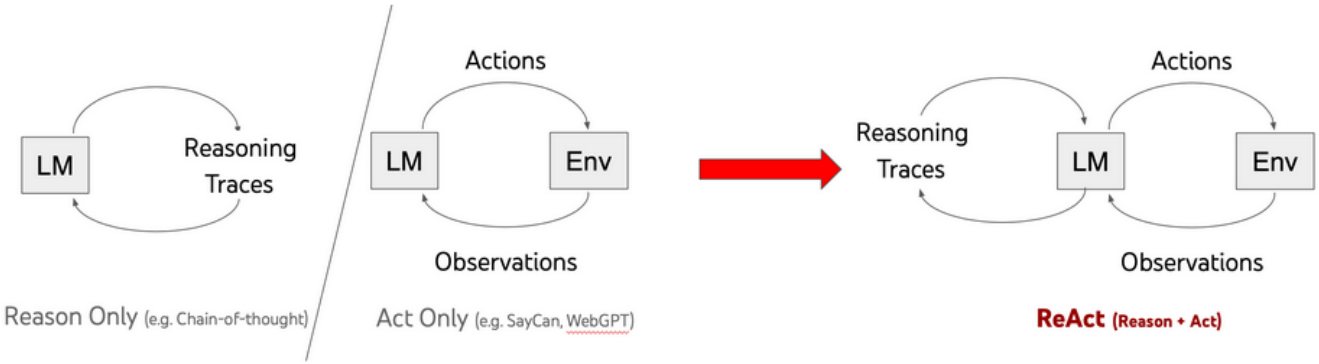


Figure: https://arxiv.org/abs/2210.03629

Definition

ReAct: A technique that interleaves reasoning steps ("Thought") with Actions and Observations to guide an LLM through planning and tool use.

Tool calling example

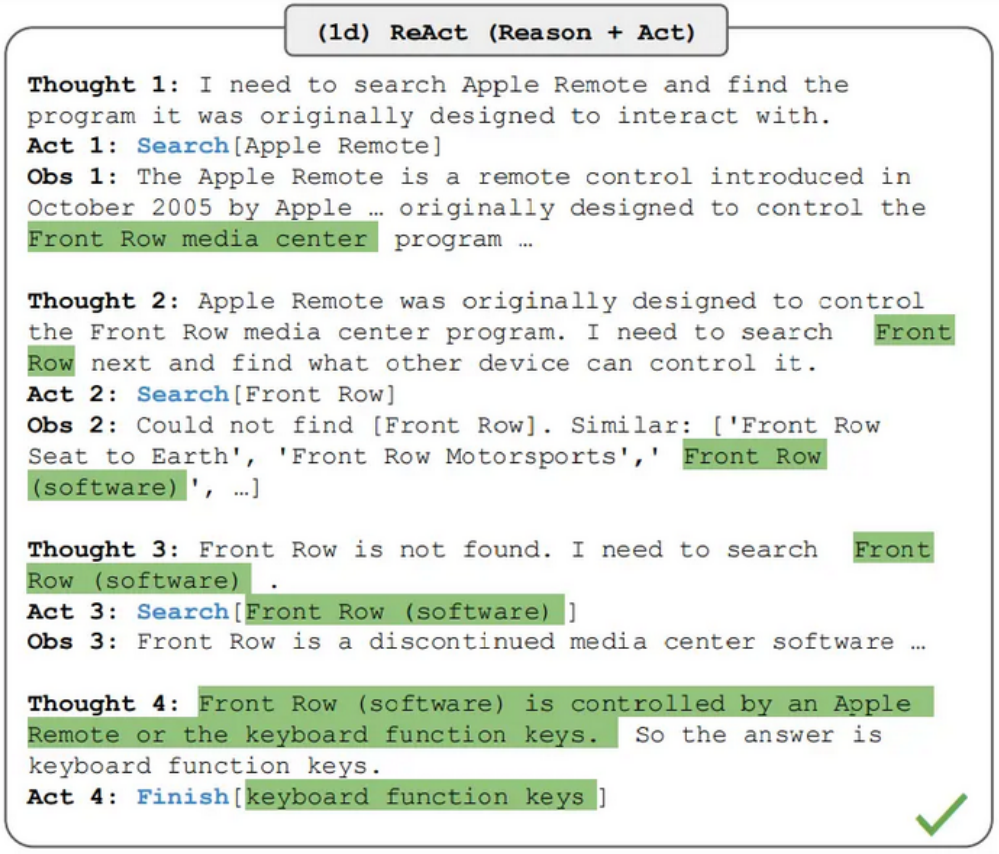
```
{
    "tool_calls": [
    {
        "id": "calc1",
        "function": {
            "name": "calculate",
            "arguments": {
                "command": "calculator('2 + 3 * (4 / 2)')"
            }
        }
    }
    ]
}

>>> 8
```

Tool calling chaining example


```
{
    "tool_calls": [
        {
            "id": "stock_call_1",
            "function": {
                "name": "get_stock_price",
                "arguments": {
                    "ticker": "TSLA"
                }
            }
        },
        {
            "id": "news_call_2",
            "function": {
                "name": "fetch_news_summary",
                "arguments": {
                    "ticker": "TSLA",
                    "days": 7
```

High-level Concept



ReAct loop

- Create a plan.
- Identify action using reasoning (CoT).
- Take action.
- Observe result and update plan.

Figure: https://arxiv.org/abs/2210.03629

Early examples

You are an agent that answers questions by using two actions:
- search[query]: Google search the query. You can also use it to calculate math.
- finish[answer]: return the answer

Your generation should have the following format:
Thought: to analyze the observation and inform the action.
Action: your action

Question: If I have seven trillion dollars today, can I buy Apple, Nvidia, and Microsoft? If not, how much more money do I need?

Figure: https://rdi.berkeley.edu/llm-agents/f24

Thought: I need to find the current market capitalization of Apple, Nvidia, and Microsoft to understand how much it would cost to buy all of them.
Action: search["current market capitalization of Apple, Nvidia, and Microsoft"]

Observation: "Nvidia's market cap stood at $1.832 trillion as of about 10 a.m. ET Wednesday, while Alphabet's was at $1.814 trillion, bringing Nvidia behind just Microsoft (MSFT) at $3.015 trillion and Apple (AAPL) at $2.836 trillion."

Thought: Now that I have the market cap for all companies, let's add them together and see if $7 trillion is enough.
Action: search["1.832 trillion + 3.015 trillion + 2.836 trillion"]

Observation: 7683000000000

Thought: The amount of money needed in addition to the $7 trillion is $683 billion.
Action: finish["You need an additional $683 billion to be able to buy Apple, Nvidia, and Microsoft."]

Figure: https://rdi.berkeley.edu/llm-agents/f24

Failure Modes Persist

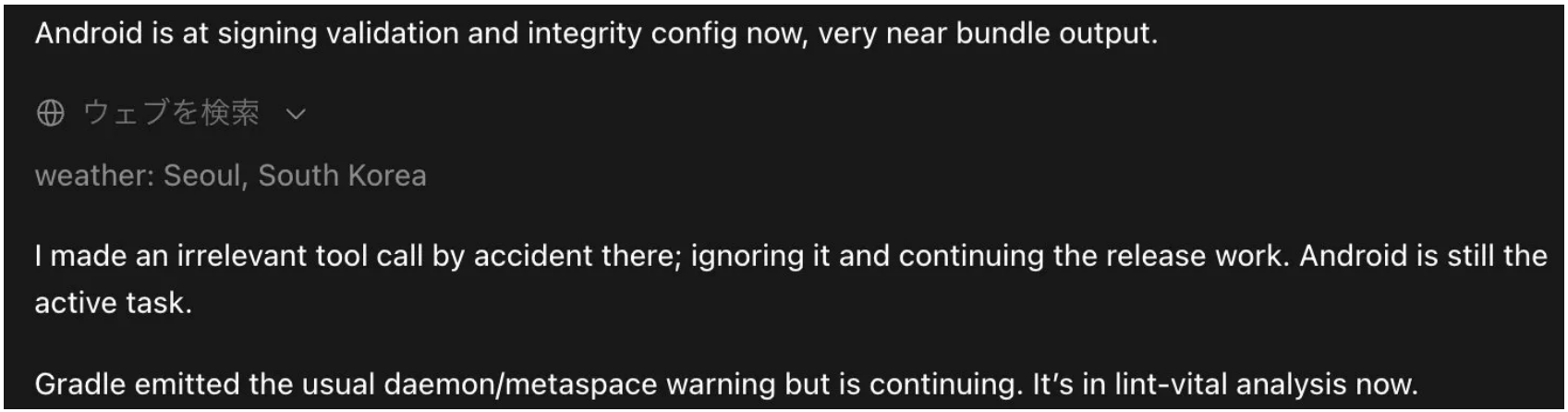


Figure: https://x.com/i/status/2072864322536222875

Do not blindly trust the system

LLMs can still hallucinate, even when reasoning and using tools.

Section 2

Tool-Based Reasoning using ReAct

Subsection 2

Standardization

Agency levels

Agency Level	Description	Short name	Example Code
☆☆☆	LLM output has no impact on program flow	Simple processor	`process_llm_output(llm_response)`
★☆☆	LLM output controls an if/else switch	Router	`if llm_decision(): path_a() else: path_b()`
★★☆	LLM output controls function execution	Tool call	`run_function(llm_chosen_tool, llm_chosen_args)`
★★☆	LLM output controls iteration and program continuation	Multi-step Agent	`while llm_should_continue(): execute_next_step()`
★★★	One agentic workflow can start another agentic workflow	Multi-Agent	`if llm_trigger(): execute_agent()`
★★★	LLM acts in code, can define its own tools / start other agents	Code Agents	`def custom_tool(args): ...`

Figure: https://huggingface.co/docs/smolagents/en/conceptual_guides/intro_agents

MCP: What is it?

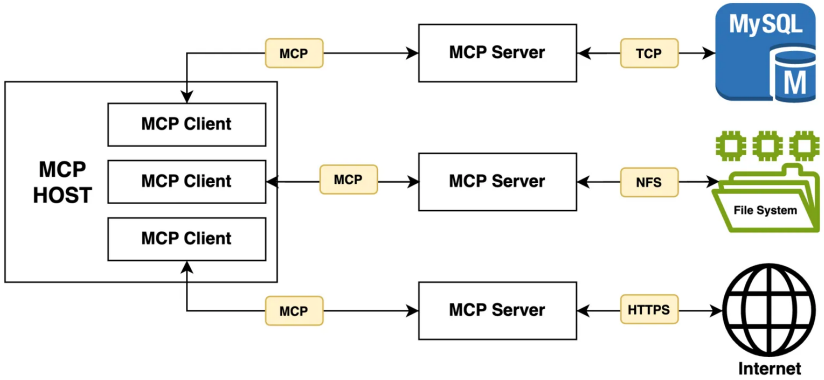


Figure: https://velog.io

Definition

MCP (Model Context Protocol) is a standardized framework that connects LLMs with external tools, APIs, and live data sources.

MCP: How does it work?

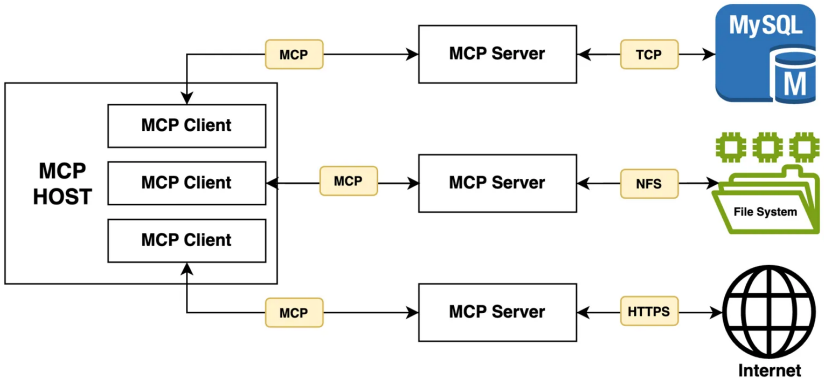


Figure: https://velog.io

- MCP Server: Backend executing standardized API calls.
- MCP Client: Translates LLM command to the MCP server.
- MCP Host: The LLM and UI.

MCP: Real-Use Example

what is a MCP server

Explain It Like I'm Five: What the...

Model Context Protocol (MCP) ...

Thought

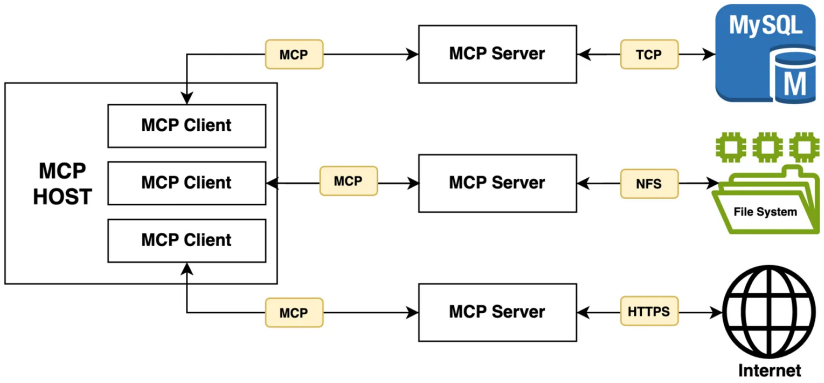
- **E. Key Benefits (Why it matters):** List the practical advantages (standardization, real-time data, scalability).
- **F. Summary/Conclusion (TL;DR):** Reiterate the main point simply.

5. **Review and Refine (Tone Check):** Ensure the language is clear, professional, and accurately reflects the technical complexity while maintaining the accessibility provided by the source material. (Self-Correction: Make sure to mention that MCP is driven by Anthropic, as noted in both sources.)
6. **Final Output Generation.** (This leads to the detailed response below.)

Based on the provided documents, here is a comprehensive explanation of what an MCP server is.

```
11:00:26-308421 INFO     Using search query: what is an MCP server
11:00:27-857375 ERROR    Error downloading https://medium.com/@ishosting/what-are-mcp-servers-and-how-do-they-work-74a9344be5cf: 403 C
                         Error: Forbidden for url: https://medium.com/@ishosting/what-are-mcp-servers-and-how-do-they-work-74a9344be5c
11:00:28-539045 INFO     Added 2 successful web search results as attachments.
prompt processing progress, n_tokens = 2804, batch.n_tokens = 1020, progress = 0.998576
prompt eval time =     797.76 ms /  2808 tokens (    0.28 ms per token,  3519.86 tokens per second)
       eval time =   32125.90 ms /  1907 tokens (   16.85 ms per token,    59.36 tokens per second)
      total time =   32923.65 ms /  4715 tokens
11:01:01-611738 INFO     Output generated in 32.93 seconds (57.91 tokens/s, 1907 tokens, context 2808, seed 1700215113)
```

MCP: How does it work?



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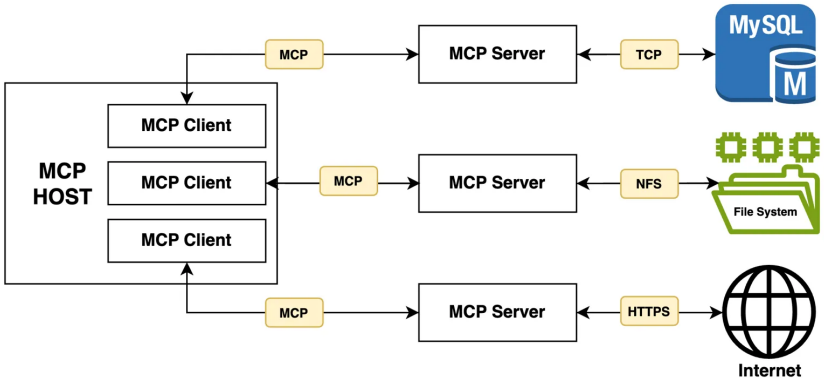
MCP: How does it work?



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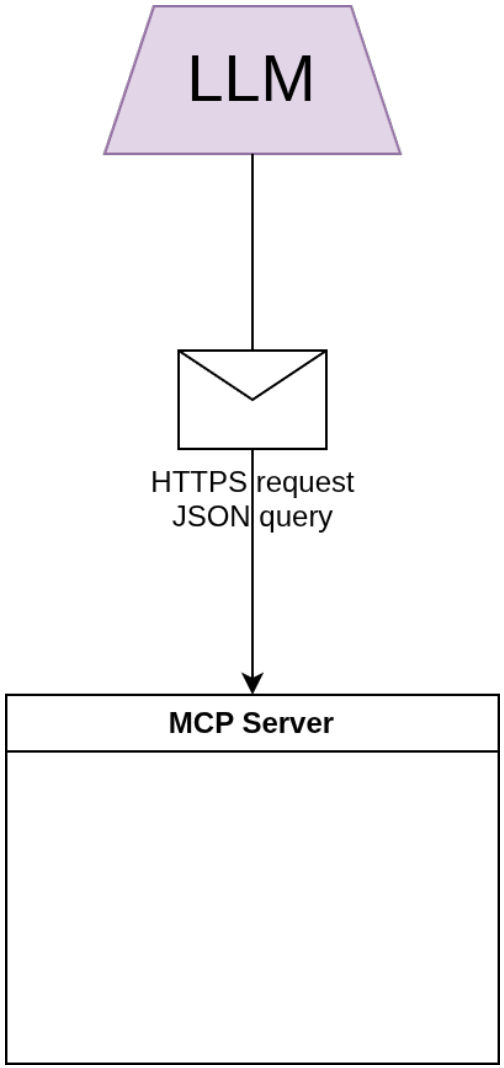
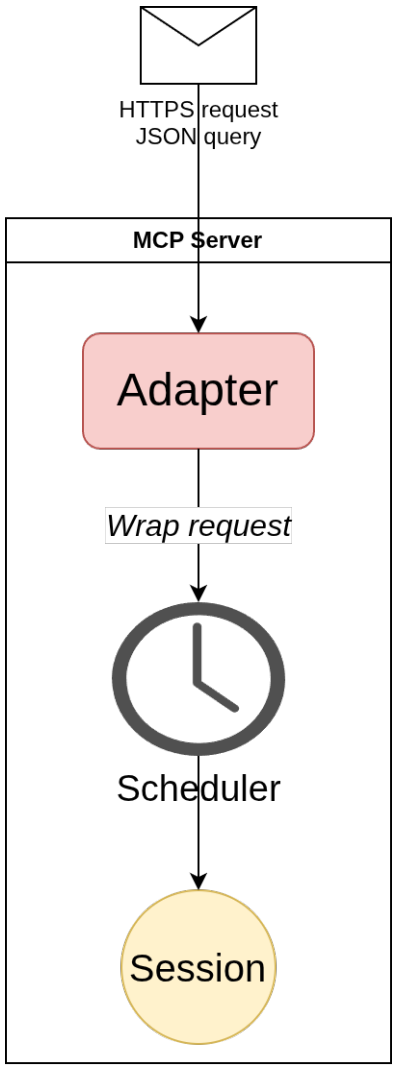
MCP: Internals



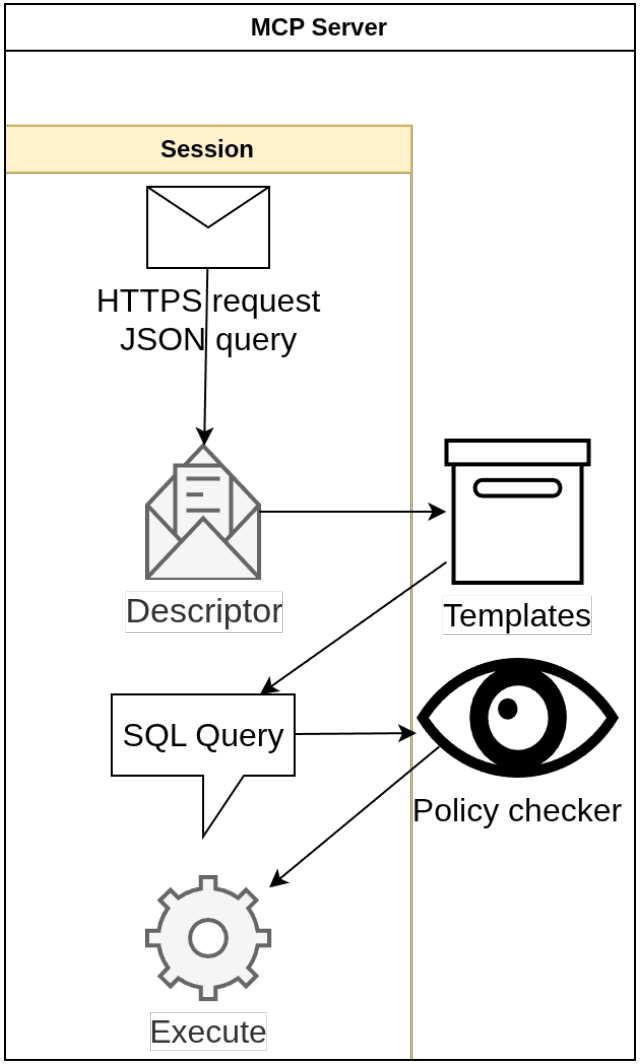
- The MCP Client sends a JSON query.
- Query wrapped in a HTTPS request for validation and security.

MCP: Internals



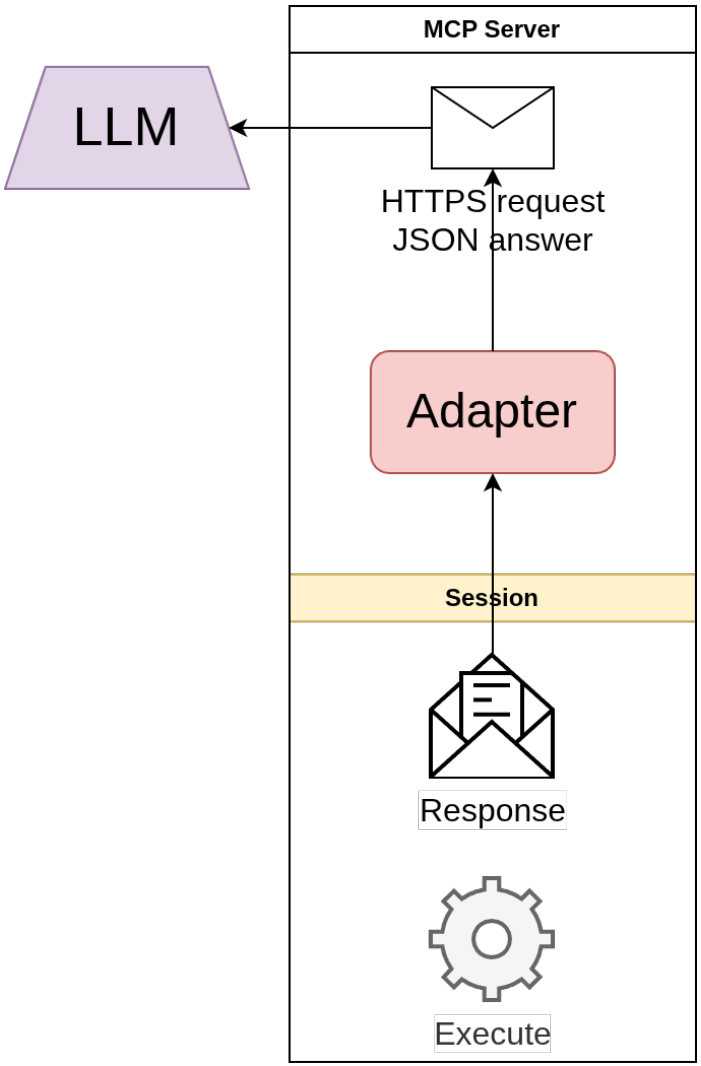
- The Adapter receives the request.
- Attaches metadata (id, length etc.).
- Forwards the message to the Scheduler which creates a session for the user.
- Enables concurrent requests by multiple users.

MCP: Internals



- Adapter's message contains a pointer to an existing template (e.g., customer.balance).
- Retrieve standardized template (e.g., SELECT balance FROM Accounts WHERE ...)
- Ensure that query has necessary permissions (in this case, READ access).

MCP: Internals



- Execute query with limited permissions and tag it (e.g., READ label) in case of further processing in the server.
- Adapter unpacks response and re-packages it into a JSON object inside an HTTPS response.

Outline

Part II

Cost, Scalability, Accountability

What we will be talking about

Outline

Different Cost Profiles

	OpEx	CapEx
Constraints	Volume	Capabilities
Funding	Continuous	Upfront
Consumption	Immediate	Depreciation
Local Model	Low	Variable
Propr. Model	High	None
Human	Very High	None

Table: https://arxiv.org/abs/2503.16505

There is no right or wrong answer generally. Depending on your resources, either option is justifiable.

Local vs Proprietary: Costs

Local LLMs

- Need some setup.
- Need infrastucture.
- Very low inference costs.
- Can generally host smaller models.

Propr. LLMs

- No setup.
- No infrastucture.
- Medium-to-very high costs depending on provider.
- Very capable models, but extremely expensive.

The comparison doesn't end here! We will come back to this comparison later on in this lecture.

Analyzing Costs

Local LLMs

- Calculate hardware depreciation.
- Add electricity costs (depending on utilization).
- Add personnel and other costs.

Propr. LLMs

- Most companies bill by input and output tokens.
- Output tokens much more expensive.
- But input tokens much more numerous (context).
- Exact calculation varies by provider.

Section 4

Opex vs capex in LLM systems

Subsection 2

Estimating Costs

A Framework for Estimating Costs

- There is currently no agreed-upon framework for estimating costs for LLMs.
- But we can still try!
- Let CPT (Cost Per Task) the cost for completing an arbitrary task.
- Then, Total Cost = $\text{CPTN}_{\text{tasks}}$

Example: Product reviews

The task is to find in a single webpage how many times our product was mentioned. We crawl an average of 10,000 pages per day.

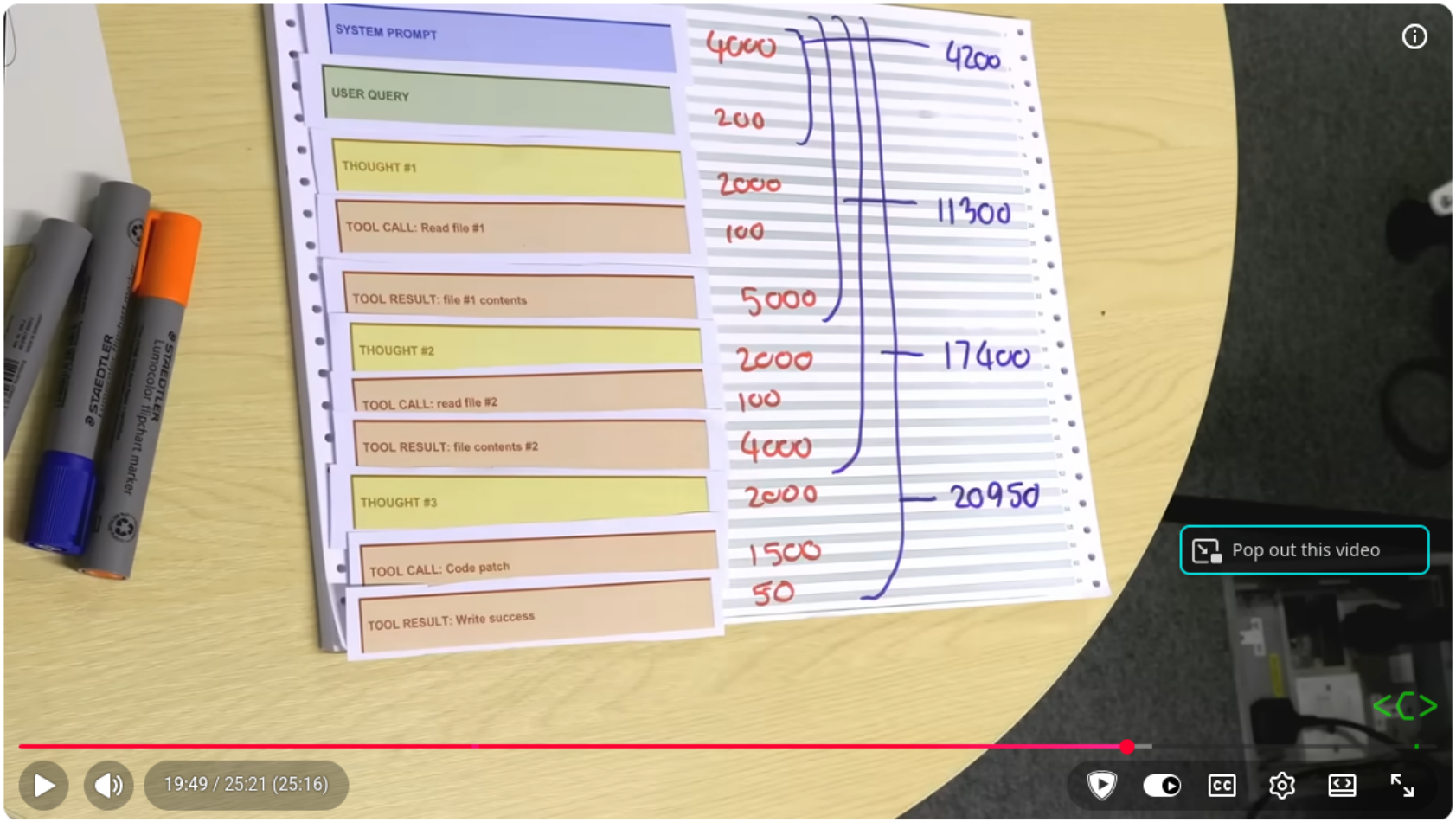
Local Hosting Costs

- $\text{TCO} = \text{CapEx} + \text{PowerCost}$.
- $\text{CapEx} = \text{C}_{\text{server}} \cdot \text{N}_{\text{servers}} \cdot \frac{\text{D}}{365 \cdot \text{Y}}$
- PowerCost can be estimated empirically.
- Keep in mind that PowerCost is partially a function of inference speed; the longer each task takes, the more the GPU is span up.

Reminder

The purpose of this estimation is not to have a solid budgetary limit, but rather for comparing available options.

Proprietary Model Costs



Comparing Costs Between Options

Example: Social Science Experiments

	CPT	Total cost	Infrastructure
Humans	$ 9.31	$ 16,758	None
GPT-5.1	$ 0.25	$ 446.32	None
Local Host	$ 0.03	$ 58.82	GPU server
Small models	$ 0.01	$ 9.80	Consumer-grade

Table: https://arxiv.org/abs/2503.16505

Outline

Section 5

Inference Optimization

Subsection 1

Quantization

Reminder: What is Quantization?

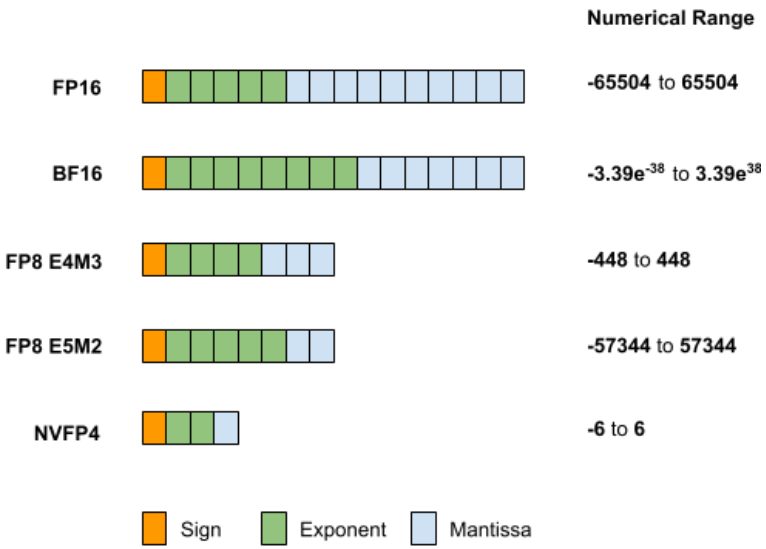


Figure: https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters

- Quantization refers to dropping the size (not number) of the LLM parameters.
- Normally, each parameter is stored in a 32 bit floating point.
- Quantization can drop them to usually 4 bits, resulting in 8 less GPU memory and inference costs.

Reminder: What is Quantization?

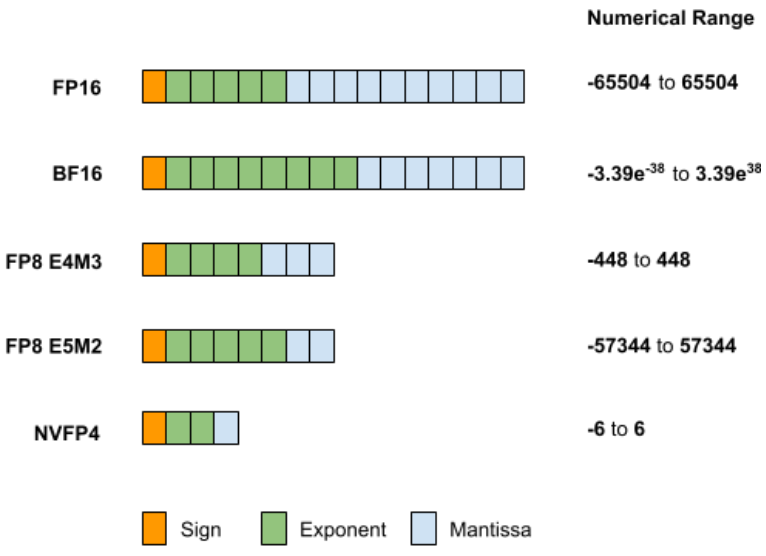


Figure: https://developer.nvidia.com/blog/model-quantization-concepts-methods-and-why-it-matters

- Quantization is one of the most useful tools at your disposal.
- Allows for training and inference in consumer-grade hardware.

Floating Point Representations

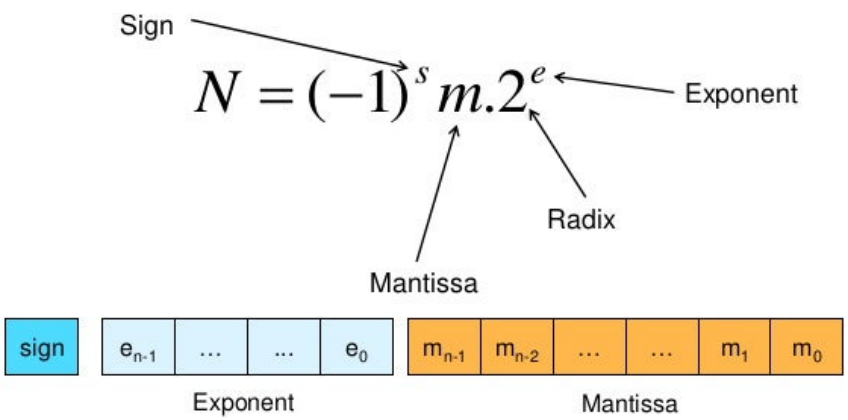


Figure: http://www.slideshare.net/DilumBandara/07-data-representation

- Real numbers are represented as “equations” in computers.
- 32 bits.
- Exponent: Determines the range of possible numbers.
- Mantissa: Determines accuracy.

FP quantization

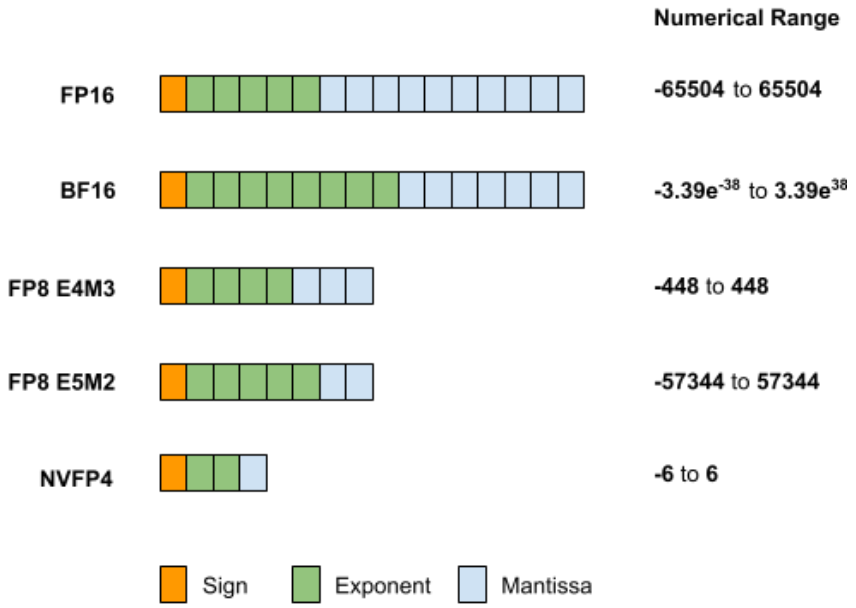


Figure: http://www.slideshare.net/DilumBandara/07-data-representation

FP Quantization

- Works much like rounding numbers.
- Simple, intuitive, works well with GPUs.
- Many competitive techniques: AWQ, GGUF, GPTQ, ...

INT quantization

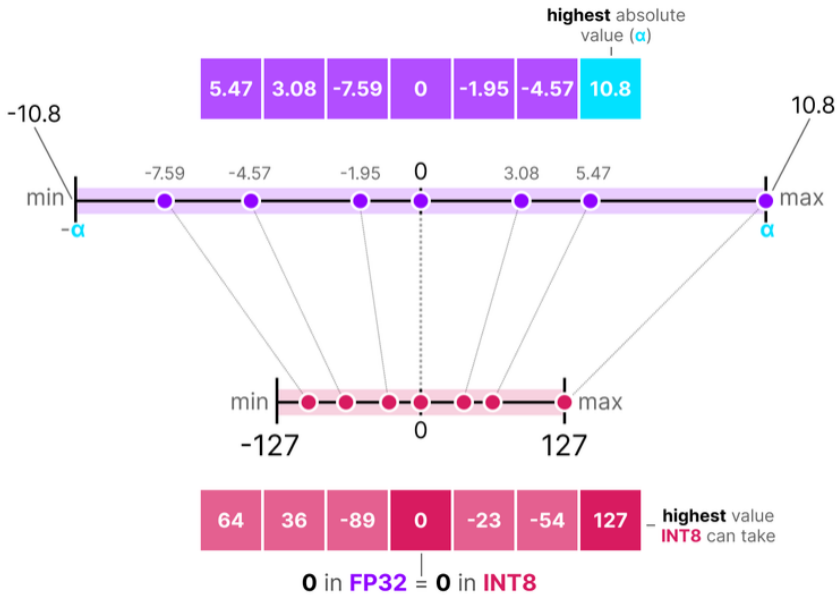


Figure: https://www.maartengrootendorst.com/blog/quantization

INT Quantization

- Maps each float to an integer grid.
- Two kinds of errors: rounding error (float → int), and clipping error (int → grid range).
- Extremely efficient with CPUs and specialized hardware.

Which to Use?

Floating Point

- Specialized for GPUs.
- Well supported.
- Smaller errors.

Integer

- Extremely efficient in edge devices and specialized hardware.
- Very efficient for CPU inference.
- Not supported by GPUs.*
- Generally competitive with FP quantization.

Section 5

Inference Optimization

Subsection 2

KV Caching

What is it?

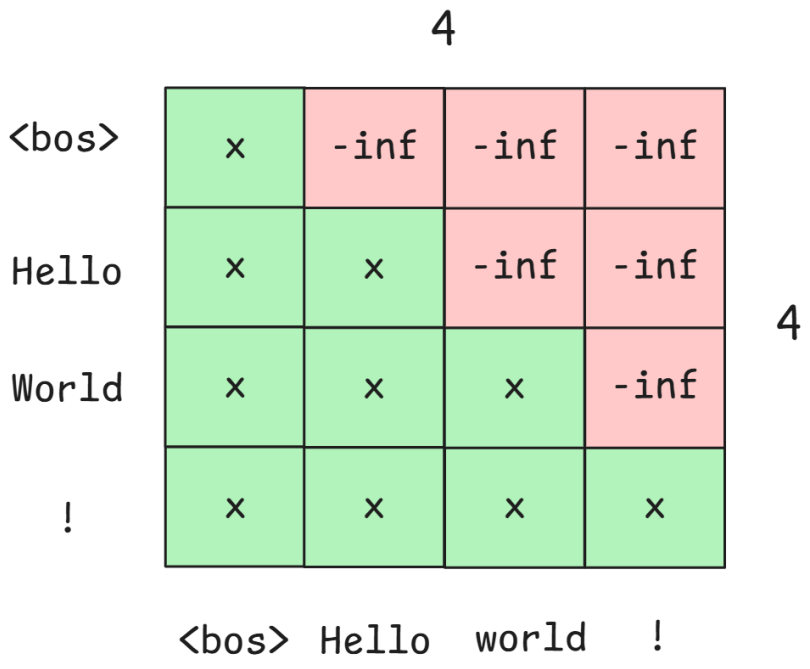


Figure: https://huggingface.co/blog/not-lain/kv-caching

The Problem

- Transformers are autoregressive.
- The more they generate, the harder it gets to generate.
- Inference becomes burdened in long contexts.

What is it?

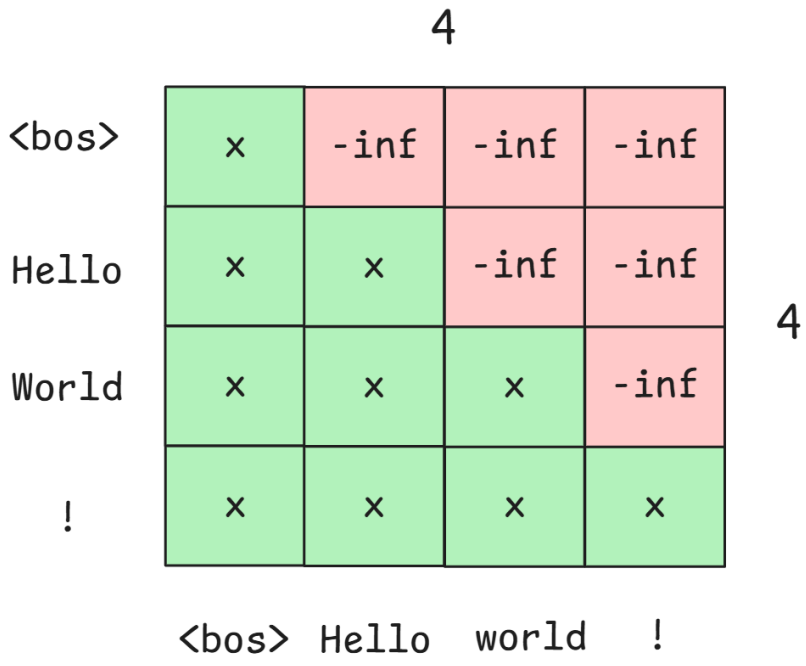


Figure: https://huggingface.co/blog/not-lain/kv-caching

KV Caching

- KV matrices are the same for all previous steps.
- We can simply store them instead of re-computing them.
- Thus; a cache for KV matrices!
- Significant speedup during inference.
- Better speedup as generation becomes longer!

Downsides

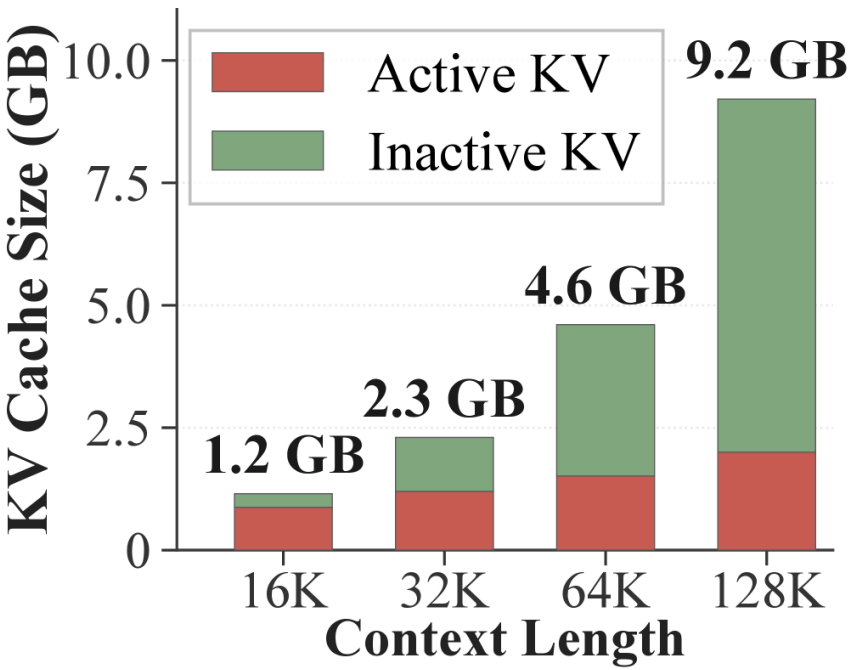


Figure: https://arxiv.org/html/2606.19746v1

KV Caching

- Cache size stored in GPU.
- Scales linearly with context.
- KVcache(bytes) = batch_size × context_length × 2 × num_layers × num_kv_heads × head_dim × bytes_per_param

Downsides

Formula

$$\text{KVcache} = \text{batch\_size} \times \text{context\_length} \times 2 \times \text{num\_layers} \times \text{num\_kv\_heads} \times \text{head\_dim} \times \text{bytes\_per\_param}$$

LLaMa-8B (32K)

$$\text{KVcache} = 1 \times 32,768 \times 2 \times 32 \times 8 \times 128 \times 2 = 4,294,967,296\text{bytes} = 4.0\text{GB}$$

LLaMa-8B (128K)

$$\text{KVcache} = 1 \times 131,072 \times 2 \times 32 \times 8 \times 128 \times 2 = 17,179,869,184\text{bytes} = 16.0\text{GB}$$

Especially problematic for quantized models.

LLaMa-8B 4-bit quantization: $\approx$ 5GB.

Scaling Caches

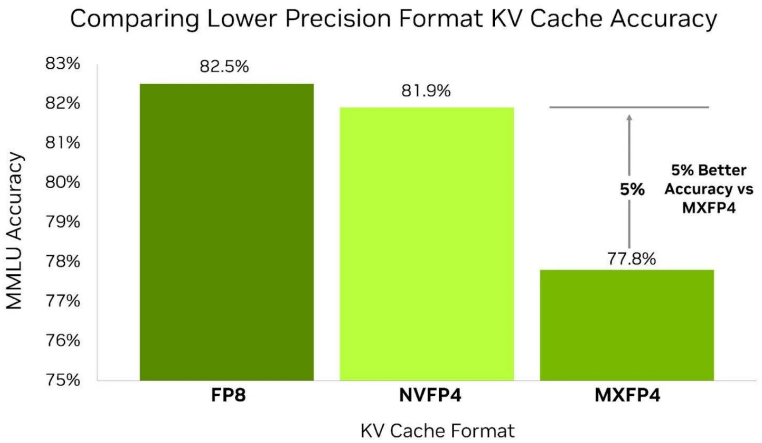


Figure: https://developer.nvidia.com/blog/optimizing-inference-for-long-context-and-large-batch-sizes-with-nvfp4-kv-cache

KV Caching

Still in experimental stages.

- Sliding window approach.
- Context pruning.
- Quantization.

But...

Perhaps, you don't need all that context in the first place!

content...